Samuel Granados

Monterrey, N.L. | +52 642 145 1046 | linkedin.com/in/samuel-granados | samu0805.github.io/portfolio | a00228152@tec.mx

Education

Instituto Tecnológico y de Estudios Superiores de Monterrey — **Monterrey, N.L.**
B.S. in Electronics Engineering — *Expected Graduation: December 2027*

- GPA 3.8/4.0 | Full Academic Leadership Scholarship (100%)
- Specialization: Power Electronics & Hardware Design

Hong Kong University of Science and Technology (HKUST) — **Hong Kong**
Exchange Program – Coursework: IC Design, FPGA, Transnational Legal Issues & Dispute Settlement — *Fall 2026*

- Started early-stage power electronics research in Prof. Wang Ping's lab.

Work Experience

Carrier — **Santa Catarina, N.L.**
Electronics Validation & Automation Intern | Electrical CoE — *June 2025 – March 2026*

- **Multiplexer:** Designed an RS-485 protocol multiplexer (ATmega328 + DG409) on a custom Altium PCB; built a C# GUI to evaluate real-time data from boards sharing the same address.
- **RCA & Validation:** Conducted Root Cause Analysis on PCB failures unresolved for over a year, proposing fixes; ran component validation testing (environmental chambers, HIL rigs) for thermal shocks, deploying results via a C# interface into Power BI.

Independent Test & Validation Intern | Software & Controls CoE — *March 2026 – Present*

- **Small Sim:** Designed an Altium shield for an STM32H563ZI running FreeRTOS, replicating residential 24VAC/PWM sensor inputs to validate embedded firmware.

Co-curricular Activities & Achievements

RoBorregos — **Monterrey, N.L.**
Director of Finance — *August 2025 – Present*

- Led a 10-member team in fundraising and budget planning, securing $400K+ MXN in corporate sponsorships and managing a $700K+ MXN total budget for national and international competitions, including RoboCup 2026 and LARC 2025.

Electronics PM – @HOME League — *June 2025 – Present*
Mexican Tournament of Robotics and RoboCup 2026

- Led the power electronics design for a 1200W power distribution and battery charging system; modified and integrated a custom BMS and distribution board within an 8-engineer team, achieving **1st Place Nationally** (April 2026).

Electronics Member | Rescue Maze League — *November 2024 – May 2025*
Mexican Tournament of Robotics

- Designed electronics under strict size/energy constraints, contributing to **2nd Place Nationally** (May 2025).

PLEI Asia Mission — **International**
Selected Mission Member — *June 2026 – August 2026*

- Selected as the first fully sponsored student to join an international business and leadership mission, conducting market research across East and Southeast Asia (China, Taiwan, Philippines, Indonesia, Singapore, Thailand, Malaysia, and Vietnam).
- Led supplier sourcing for DYL Medic, evaluating 30+ potential mobility-aid suppliers (6 visited on-site), and studied a leading shelter/nearshoring agency's Philippines model for Intugo – the market assigned for that engagement – to inform future collaboration.

EcoShell Tec Racing — **Monterrey, N.L.**
Electronics Team Member — *August 2024 – June 2025*

- Built ESP32-based vehicle telemetry (UART/I2C) for real-time status acquisition and post-run analysis in MATLAB.

Engineering Projects

Power Electronics

- **Variable Frequency Drive (VFD) – AMI Automation:** Designed and implemented a two-level Voltage Source Inverter (VSI) driving a 3-phase induction motor via scalar V/f control; developed sinusoidal PWM (S-PWM) firmware in C (STM32, HAL) generating three 120°-phased references at a 100µs sample rate, validated across a 9–36Hz operating range with inverter output up to 109 V_{RMS}.
- **Bendix:** Designed and simulated a custom inductor using COMSOL Multiphysics, optimizing magnetic performance and validating design constraints in collaboration with industry engineers.

Embedded Systems & Controls

- **OPmobility Challenge 2025:** Designed and validated an ECU diagnostic prototype for automotive LED lighting on an ATmega328P (Arduino UNO); implemented ADC-based fault detection (overvoltage, undervoltage, open/short circuit, overtemperature via thermistor), full-duplex UART reporting, and persistent EEPROM fault logging, verified through unit and integration testing.
- **Acuity Brands:** Designed an op-amp analog signal-conditioning circuit for a motor-driven window blind (gain, filtering, stability) without embedded firmware.
- **DIRAM:** Developed and tuned a PID-based position control system, analyzing system response to meet industrial performance criteria.

Skills and Interests

Programming & Control: C++, C#, C, Python, MATLAB, Simulink.

Power Electronics: Voltage Source Inverters, Scalar V/f & S-PWM Motor Control, Power Converters, Magnetics Design (COMSOL).

Embedded Systems: Microcontrollers (ESP32, AVR/ATmega, PIC18, TI F28379D, STM32), Data Acquisition (DAQ) Systems, Verilog, FPGA, IC Design.

Communication Protocols: CAN, RS-485, I2C, UART, TCP/IP, SCPI.

Design: Hardware design (Altium, KiCad, OrCad), Simulation (LTspice, Multisim).

Electronics: Analog IC Design Fundamentals, ADC/DAC Principles, Semiconductor Devices, Optoelectronics, Amplifiers, Telecommunications, Signal Processing.

Languages: English (B2 - 6.5 IELTS), Spanish (Native).

Certifications: Google Data Analytics (2025). Lean Six Sigma Green Belt (2024). MIT NanoLab - Intro to Micro and Nano Production (2024).